

# Gabriel Eze

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## EDUCATION

**B.Sc. Data Science & Mathematics**, Centre College

08/22 – 05/26 | Danville, USA

Relevant Coursework: **Abstract Algebra, Advanced Interpolation Methods, Applied Machine Learning, Bayesian Data Analysis, Building AI-Powered Applications, Data Manipulation & Visualization, Linear Algebra I & II, Mathematical Statistics, Probability Theory, Python Programming and Problem Solving, Real Analysis, Statistical Modeling.**

## RESEARCH

**Stack Domination Density of Hypercubes**, Graph Theory

05/25 – 08/25 | Dr. Brauch, Dr. Wiglesworth

- Formulated and proved results for stack domination density in  $C_m \times P_2 \times P_n$ , deriving bounds across residue classes of  $m$ .
- Designed modular “Clockwise” and “Domination” algorithms to construct minimal dominating sets and validate proofs through large-scale computation.
- Co-authored a 38-page research manuscript integrating combinatorial proofs and algorithmic verification for potential undergraduate publication.

**Pitch Modeling**, Carnegie Mellon SURE Camp

06/24 – 07/24 | Quang Nguyen, Dr. Yurko

- Conducted advanced statistical analysis and modeling to predict pitch effectiveness (Stuff+) for MLB players, leveraging large datasets from FanGraphs and Baseball Savant.
- Applied Random Forest, Lasso regression, and HGBR models to analyze the complete arsenal of pitches per pitcher, enhancing predictive accuracy by modeling multiple pitch types and providing competitive RMSE results.
- Developed a pitch recommendation tool that provided actionable insights for MLB teams, optimizing player performance by suggesting ideal pitch sequencing based on past game data.

*In-person presentation @ CMSAC 24—sole on-site team*

**Sports Analytics**

05/23 – 07/23 | Dr. Heath, Dr. Poudel

- Utilized Python to lead data cleaning and parsing efforts within a complex big data pipeline, ensuring data accuracy and reliability for basketball game analysis.
- Tracked possession data and employed advanced statistical techniques to assess player substitutions, providing vital insights for informed coaching decisions and game outcome analysis.
- Collaborated on the development of win probability models based on real-time game data, empowering the coaching staff with actionable insights for strategic game planning.

*Elevator pitch + poster presentation @ CMSAC 23*

## PERSONAL PROJECTS

**Forest Fire Simulation**

11/22 – 12/22

Designed class-based forest simulation modeling state transitions on a  $10 \times 15$  grid.

**DNA Sequence Analysis**

10/22 – 11/22

Developed sliding-window STR detection algorithm to quantify repeat sequence from genomic input.

## SKILLS

**Computing** — Harmonic Analysis, Logic, Q-Ball Imaging

**LLM** — Chunking, Doc2Query, Embeddings, Hit@K, Prompt Engineering, Red-Team Testing, Tool Use

**Programming** — MATLAB, Python, R, SQL

**Tools** — ChromaDB, Endpoints, Git, JupyterLab, LaTeX, MySQL, Pydantic, Quarto, Regex, Tableau, VS Code

## AWARDS

**Best Visualization**, ASA DataFest @ EKU

2025

Analyzed national office leasing data to deliver strategic market insights for Savills in 3-day competition.